SVNIT · DEPARTMENT OF CIVIL ENGINEERING

SafeRoute Gujarat

Speaker guide for the 15-minute project presentation, slide by slide

Team: Chinta Lakshya (Civil) · Damerla Akshara (Civil) · Arnav Balodi (CSE) · Veer Kishan Whabi (ECE)

Supervisor: Prof. Shriniwas Arkatkar

How to use this guide: find your slides in the table on the next page. For each slide, read “Understand it first” so you can answer questions, then practise the script aloud with a timer. The scripts are written to fit the time shown; the speaker notes inside the .pptx are longer reference versions.

Who presents what

Veer Veer Kishan Whabi (ECE)

Role: UI/UX and backend
Slides: 1, 2, 12, 13 · **Speaking time:** 3 min 00 s
Know well: The interface features, the API endpoints and input validation, how the pieces connect, and the live demo.

Akshara Damerla Akshara (Civil)

Role: Theory and data sources
Slides: 3, 4, 7, 8, 16 · **Speaking time:** 4 min 30 s
Know well: MoRTH figures, every row of the evidence table, the five data sources and blackspots, and the four safety formulas.

Chinta Chinta Lakshya (Civil)

Role: Survey and field data
Slides: 5, 6, 9, 14, 17 · **Speaking time:** 4 min 00 s
Know well: All survey numbers with their n, the six traffic-flow and scoring formulas, and the SVNIT → Station case study.

Arnav Arnav Balodi (CSE)

Role: Risk modelling and routing
Slides: 10, 11, 15 · **Speaking time:** 3 min 00 s
Know well: The graph and risk model, why it is not ML, A* and the admissible heuristic, the cost formula, detour cap, Pareto front, and the validation numbers.

#	Slide	Speaker	Time	Running
1	SafeRoute Gujarat (title)	Veer	20 s	0:00–0:20
2	Team contributions	Veer	50 s	0:20–1:10
3	Why a safety-first route	Akshara	50 s	1:10–2:00
4	What the evidence says about road risk	Akshara	1 min	2:00–3:00
5	Commuter survey: who answered	Chinta	50 s	3:00–3:50
6	What commuters told us	Chinta	55 s	3:50–4:45
7	Data we collected	Akshara	50 s	4:45–5:35
8	Safety formulas used in the model	Akshara	1 min 10 s	5:35–6:45
9	Traffic-flow and scoring formulas	Chinta	1 min	6:45–7:45
10	The risk model, step by step	Arnav	1 min	7:45–8:45
11	Routing engine: turn-aware A* search	Arnav	1 min 10 s	8:45–9:55
12	System architecture	Veer	50 s	9:55–10:45
13	The interface	Veer	1 min	10:45–11:45
14	Case study: SVNIT → Surat Station	Chinta	55 s	11:45–12:40

15	Validation: 400 simulated trips	Arnav	50 s	12:40–13:30
16	Limitations and next steps	Akshara	40 s	13:30–14:10
17	Thank you	Chinta	20 s	14:10–14:30

Total about 14 min 30 s, leaving a little room before questions. Hand over with a short phrase like “next, the data we used”. Don't say the next speaker's name.

Where the formulas come from

The **ideas and priorities** come from the road-safety literature and data (HSM, Indo-HCM, MoRTH). The **formulas** are standard published traffic-engineering results or our own design, and some **parameter values** are our assumptions. Say this accurately if asked.

Item	Source	Status
Crash factors; roadway factors in about a third of crashes	AASHTO Highway Safety Manual	Motivation and design choices
Right turns conflict with through traffic	Indo-HCM (2017) conflict studies	Concept from the studies; factor values 0.75 / 1.6 / 2.2 are our assumptions
Vulnerable road users: 48% of deaths	MoRTH	Motivation for two-wheeler and walking modes
40 km/h reference speed	Typical urban posted limit	Our choice of reference; the severity formula itself is Nilsson
Severity power model S(v)	Nilsson (2004), standard road-safety literature	Standard result
Lighting factor 1/0.72, signal 0.70	AASHTO Highway Safety Manual (CMFs)	Values from the HSM literature
Bridge ×1.25	Derived from MoRTH 2022 Table 6.9 (3.0% crashes, 4.6% deaths)	Our estimate
Divided ×0.80, unpaved ×1.30, roundabout 0.60	Order-of-magnitude values from HSM-style literature	Assumptions
Empirical Bayes	Hauer (1997); used in the HSM	Over-dispersion 0.6 is an assumed typical value
BPR travel time	Bureau of Public Roads (1964)	Standard formula
Webster signal delay	Webster (1958)	Standard formula
Time-of-day risk	MoRTH 2022 Table 7.3 (crash shares)	Traffic shares per hour are assumed
Pedestrian fatality P(v)	Rosén & Sander (2009)	Standard result
Calibration constant k; safety score	Our own design	Calibrated to MoRTH 2022 city totals

Safe phrasing: “The concept comes from [HSM / Indo-HCM / MoRTH]; the formula is a standard result from [source]; the parameter value is our assumption, calibrated so the city totals match MoRTH.”

Things we must not claim

- That the routes are **proven** safer. Risk is modelled; no observed crash data on these exact routes.
- That we use **machine learning**. It is a calibrated statistical model.
- That we use **live traffic** or **AQI**. Travel time uses an assumed hourly demand profile; AQI belongs to another team.
- That the survey represents **all of Surat**. It represents mostly young SVNIT commuters.

Slide-by-slide guide

1 SafeRoute Gujarat (title)

Veer 20 s · running time 0:00–0:20

ON THE SLIDE

- Project name, one-line description, SVNIT Civil Engineering
- Team names with branch and role; supervisor Prof. Shriniwas Arkatkar
- Screenshot of the working prototype (Surat, 9 pm, dark mode)

UNDERSTAND IT FIRST

The original proposal ("Route Optimization Based on Safety, AQI & Efficiency") had three objectives. AQI and efficiency went to other teams; ours is **safety**. Time and distance still appear, but only so a safe route can be compared fairly with the normal fastest route.

The screenshot is real output from our app, not a mock-up.

SCRIPT (fits 20 s)

Good morning. We are presenting SafeRoute Gujarat, our mini project under Prof. Arkatkar. Of the three objectives in the original proposal, our team owns safety.

In one line: Google Maps shows you the fastest route; our system shows that route next to a safer one, and explains why it is safer and what it costs in minutes.

IF ASKED

Why only safety?
AQI and efficiency were assigned to other teams. Our model can later be combined with theirs; that is listed in next steps.

2 Team contributions

Veer 50 s · running time 0:20–1:10

ON THE SLIDE

- Four cards, one per member: name, branch, role and three things each person did

UNDERSTAND IT FIRST

The professor asked that each member's contribution be clear. Each person should be able to explain their own card in detail if questioned.

SCRIPT (fits 50 s)

Here is how we split the work. Akshara, from Civil, handled the theory: the literature review covering the Highway Safety Manual, Indo-HCM and the MoRTH reports. She also collected the crash statistics and the 22 accident blackspots.

Chinta, also Civil, designed and ran our commuter survey with 100 responses and worked on the traffic-flow formulas. Arnav, from CSE, built the risk model, its calibration, the A* routing engine and the validation. I am from ECE; I built the map interface, route comparison and charts, and the API layer that connects them.

We go in that order: problem and data, the model, the system, then results.

3 Why a safety-first route

Akshara 50 s · running time 1:10–2:00

ON THE SLIDE

- Bar chart: 2022 road accidents and deaths for Ahmedabad, Surat, Rajkot, Vadodara
- Three callouts: 1,112 deaths in a year; 20.4% of crashes between 6 and 9 pm; no mainstream navigation app ranks routes by safety

UNDERSTAND IT FIRST

All numbers are from MoRTH, *Road Accidents in India 2022*. Table 6.2 gives the million-plus cities: Ahmedabad 1,711 accidents / 488 killed, Surat 683 / 293, Rajkot 498 / 150, Vadodara 478 / 181. 488 + 293 + 150 + 181 = 1,112.

Gandhinagar is not a million-plus city, so it has no separate row in MoRTH; it is modelled together with Ahmedabad.

Severity: Surat has 293 deaths per 683 crashes, about 43 per 100, against roughly 37 nationally. The 20.4% figure is MoRTH Table 7.3 (18:00–21:00), the worst 3-hour window.

Key message: crashes come from factors that can be anticipated and altered, so they are predictable and preventable.

SCRIPT (fits 50 s)

> Why safety? These are MoRTH's official 2022 figures. Ahmedabad had 1,711 road accidents and 488 deaths, Surat 683 and 293, Rajkot 498 and 150, Vadodara 478 and 181: 1,112 deaths in one year in our four cities.
>
> Surat is especially severe, with about 43 deaths per 100 crashes against a national average of about 37. Timing matters too: 20.4% of India's crashes happen between 6 and 9 pm.
>
> Crashes are predictable and preventable. Yet no mainstream navigation app ranks routes by safety. That is the gap we address.

IF ASKED

Why is Gandhinagar missing from the chart?
MoRTH publishes city rows only for million-plus cities. Gandhinagar is routed as part of the Ahmedabad–Gandhinagar region.

4 What the evidence says about road risk

Akshara 1 min · running time 2:00–3:00

ON THE SLIDE

- Two-column table: “Finding” (with its source) and “How we used it in the model”
- Seven rows: crash factors, right-turn conflicts, vulnerable road users, blackspots, V/C and delay, HSM factors, the 5 E's

UNDERSTAND IT FIRST

This slide is our literature review and data findings: the evidence each part of the model rests on.

Crash factors (AASHTO Highway Safety Manual): crashes come from road, environment, traffic and driver factors; roadway factors are present in about a third of crashes. We model road class and speed, lighting and time of day, and traffic exposure. Driver behaviour is **not** modelled; we have no driver data.

Right-turn conflicts (Indo-HCM 2017 conflict studies): at T-junctions the main conflict is between right-turning and through vehicles. The **idea** comes from these studies; the **numbers** we use (right 1.6×, U-turn 2.2×, left 0.75×) are our modelling assumptions.

Vulnerable road users (MoRTH): pedestrians, cyclists and two/three-wheelers are 48% of road deaths, so we have two-wheeler and walking modes. **Blackspots**: MoRTH's 500 m definition and the GRSA audits. **V/C and delay**: standard traffic-flow theory (BPR, Webster). **5 E's**: Engineering, Education, Enforcement, Emergency care, Evaluation.

SCRIPT (fits 1 min)

Before the data, this slide shows the evidence our design is built on. The Highway Safety Manual shows crashes come from road, environment and traffic factors. These became our road-class, lighting and time-of-day factors.

Indo-HCM conflict studies show the main conflict at T-junctions is between right-turning and through vehicles, so our router penalises right turns and U-turns.

MoRTH data shows vulnerable road users are 48% of deaths, so we built separate two-wheeler and walking modes. Crashes cluster at blackspots, so we weight 22 documented ones, and traffic-flow theory tells us delay rises sharply near capacity.

And from the 5 E's of road safety: our route choice is engineering, and we also show police proximity, nearest hospital and a validation study.

IF ASKED

Where does the 1.6 for right turns come from?
The idea is from Indo-HCM conflict studies; the value is our assumption, chosen so right turns cost clearly more than going straight. With conflict or crash data it could be estimated properly.

5 Commuter survey: who answered

Chinta 50 s · running time 3:00–3:50

ON THE SLIDE

- Four tiles: 100 responses, 99% SVNIT students and scholars, 97% under 30, 61/39 male/female
- Chart: main travel mode (auto 40%, car/taxi 25%, two-wheeler 20%, BRTS 15%)
- Chart: route usually taken (switch using live traffic 45%, Route A 34%, Route B 18%, Route C 3%)

UNDERSTAND IT FIRST

Google Form “What's Your Smartest Route”, 10 Oct 2026, corridor SVNIT ↔ Surat Railway Station.

Route A = Dumas Road and Ring Road; Route B = Athwa Lines and Chowk Bazaar; Route C = VIP Road / Bhatar / Udhna Darwaja.

Limitations to state openly: respondents are mostly young SVNIT students, so results describe campus commuters, not all of Surat. Mode shares are self-reported, not measured traffic volumes.

SCRIPT (fits 50 s)

To ground the project in real experience we surveyed one corridor we all use: SVNIT to Surat Railway Station. We got 100 responses: 99% SVNIT students and scholars, 97% under 30, 61% male and 39% female. So this represents young campus commuters, and we say that openly.

40% mainly use auto-rickshaws, 25% cars or cabs, 20% two-wheelers and 15% BRTS. And 45% don't stick to one route: they switch based on live traffic in Google Maps. People already let an app choose their route, so if the app considered safety, their choices would change too.

IF ASKED

Is 100 responses enough?
For understanding priorities on one corridor, yes; it is not a city-wide sample. We use it to guide the design, not to prove a route is safer.

6 What commuters told us

Chinta 55 s · running time 3:50–4:45

ON THE SLIDE

- 57.9% would take a 2–3 minute longer route if it is safer, cleaner and less congested (n = 95)
- 45.1% named rash driving or speeding near intersections as the top hazard (n = 91)
- 31% said Station Approach Road has the worst congestion (n = 100)
- 67.4% face a 5–15 minute delay (n = 95)
- Right panel: how this shaped the model (detour limit, junction risk, test corridor)

UNDERSTAND IT FIRST

Each question has its own number of answers (n), because not everyone answered every question. Always quote the percentage with its n.

Other answers: 38.9% said “depends on urgency”; hazards after speeding were traffic conflicts 23.1% and missing pedestrian crossings 15.4%; SVNIT Circle / Kargil Chowk 28% and Ring Road flyover / Sahara Darwaja 14% for congestion; shortest travel time was the top reason for choosing a route (32%).

Design links: people accept minutes, not a long detour → we cap the detour (25% by default) and offer a Balanced route. Danger is at junctions → the model costs every junction and turn. The corridor becomes our case study on slide 14.

SCRIPT (fits 55 s)

Four findings shaped the design. First, 57.9% would take a route 2 to 3 minutes longer if it were safer and cleaner, and another 38.9% said it depends on urgency. So a safety route is acceptable if the time cost is small. That is why our engine limits the detour and also offers a Balanced route.

Second, the top hazard was speeding near intersections at 45%, so our model scores every junction and every turn, not just road length. Third, Station Approach Road was the most reported congestion point. Fourth, two-thirds face 5 to 15 minutes of delay.

These are reported experiences, not measurements; the objective data comes next.

IF ASKED

Did the survey measure AQI or safety?
No. It measured preferences and experiences. Objective safety comes from MoRTH data and blackspots; AQI belongs to the other team.

7 Data we collected

Akshara 50 s · running time 4:45–5:35

ON THE SLIDE

- Table of five sources: OpenStreetMap, NASA VIIRS night lights, MoRTH Road Accidents 2022, accident blackspots, commuter survey
- Card: 22 verified blackspots (Ahmedabad 15, Vadodara 4, Gandhinagar 1, Surat 1, Rajkot 1); worst is Jambuva bridge, NH-48: 36 crashes, 30 deaths (2016–18)

UNDERSTAND IT FIRST

OpenStreetMap: the road network (23,273 km, about 2.3 lakh junctions) plus road class, one-ways, signals, police stations and hospitals.

NASA VIIRS night lights: OSM has almost no street-light data for Gujarat, so satellite night-time brightness (median of 8 clear nights, Dec 2025 – Feb 2026) estimates whether a road is lit. Resolution is about 500 m.

Blackspots: Gujarat Road Safety Authority audit of Ahmedabad (reported June 2026), State Traffic Branch and GRSA festive-season list (Oct 2025), and MoRTH's national-highway top-ten blackspot annexure (Road Accidents in India 2019, Annexure 52). Four locations that could not be pinned down reliably were dropped, not guessed.

MoRTH defines a highway blackspot as a 500 m stretch with 5 or more fatal/grievous crashes, or 10 or more deaths, in 3 years.

SCRIPT (fits 50 s)

Five data sources feed the system, all public. OpenStreetMap gives the road network: 23,273 km and about 2.3 lakh junctions, with signals, police stations and hospitals. OpenStreetMap has almost no street-light data for Gujarat, so we used NASA satellite night-lights to estimate whether a road is lit.

MoRTH 2022 gives each city's crash totals and time-of-day pattern. We collected 22 accident blackspots with real crash and death counts from the Gujarat Road Safety Authority, the State Traffic Branch and MoRTH. The worst is the Jambuva bridge on NH-48: 36 crashes and 30 deaths in three years. Sites we couldn't locate reliably were dropped, not guessed. And finally, our own survey.

IF ASKED

Why so few blackspots in Surat and Rajkot?
Only what is publicly reported with location and counts was used. Coverage depends on what authorities publish.

Is satellite light accurate per street?
No; about 500 m resolution. It separates lit city roads from dark outskirts but not neighbouring streets. Municipal streetlight data would fix this.

8 Safety formulas used in the model

Akshara 1 min 10 s · running time 5:35–6:45

ON THE SLIDE

- Four cards, each with a typeset formula: speed severity, lighting and road features, turns at junctions, crash history at blackspots

FORMULAS ON THIS SLIDE

1. Severity (Nilsson power model)

$$S(v) = (1-\varphi)\left(\frac{v}{40}\right)^2 + \varphi\left(\frac{v}{40}\right)^4, \qquad \varphi = 0.33$$

v = operating speed (km/h); φ = share of crashes that are fatal. Injury harm grows with speed squared, fatal harm with speed to the fourth. At 80 km/h: $S = 0.67 \cdot 4 + 0.33 \cdot 16 \approx 8$, i.e. about 8× the harm at 40 km/h.

2. Lighting factor (HSM crash modification factor)

$$\mathrm{CMF}_{\mathrm{light}} = 1 + 0.39\,(1 - p_{\mathrm{lit}})\,d(h)$$

p_{lit} = probability the road is lit (from satellite + road class); $d(h)$ = darkness at hour h (0 in daytime, 1 at night). 0.39 comes from 1/0.72 − 1: lighting reduces night injury crashes by about 28%.

3. Turns at junctions

$$R_{\mathrm{turn}} = k \cdot R_{\mathrm{junction}}, \qquad k_{\mathrm{left}} = 0.75,\ k_{\mathrm{straight}} = 1.0,\ k_{\mathrm{right}} = 1.6,\ k_{\mathrm{U}} = 2.2$$

4. Empirical Bayes at blackspots

$$w = \frac{1}{1 + 0.6\,N_{\mathrm{pred}}\,Y}, \qquad N_{\mathrm{EB}} = w\,N_{\mathrm{pred}} + (1-w)\,\frac{N_{\mathrm{obs}}}{Y}$$

N_{pred} = crashes per year the model predicts within 250 m of the blackspot; N_{obs} = crashes recorded over Y years. The risk multiplier is $N_{\mathrm{EB}}/N_{\mathrm{pred}}$ (kept between 1 and 8, then adjusted for deaths per crash).

UNDERSTAND IT FIRST

Why speed matters so much: doubling speed quadruples injury harm and multiplies fatal harm 16×. So a fast flyover is not automatically better just because it is quicker. This is why our safe routes often avoid high-speed arterials.

Why 40 km/h? It is the reference speed where severity = 1. It is the typical urban posted speed limit.

Empirical Bayes in plain words: if a place has few recorded crashes over a short period, we trust the model more (w close to 1). If it has a long, heavy record, we trust the record more (w close to 0). This stops one unlucky year from dominating. Example: Jambuva bridge gets ×11.3; Odhav Ring Road ×3.1.

Other factors used: bridge ×1.25 (MoRTH 2022 shows bridges have 3.0% of crashes but 4.6% of deaths), divided road ×0.80, unpaved ×1.30, signalised junction ×0.70, roundabout ×0.60.

SCRIPT (fits 1 min 10 s)

These are the exact formulas in our model. First, severity: Nilsson's power model. Injury harm grows with speed squared and fatal harm with speed to the fourth, weighted by φ = 0.33, the share of

fatal crashes from MoRTH. The reference speed of 40 km/h is the typical urban speed limit. At 80 km/h harm is about 8 times that at 40.

Second, Highway Safety Manual factors: lighting cuts night crashes by 28%, so an unlit road at night costs up to 1.39 times more. Bridges get 1.25. Third, turns: a right turn crosses opposing traffic, so it costs 1.6 times a straight move, and a U-turn 2.2.

Fourth, Empirical Bayes blends a blackspot's recorded crashes with the model's prediction, so one bad year cannot dominate. The Jambuva bridge gets an 11-fold weight.

IF ASKED

Where does φ = 0.33 come from?
MoRTH 2022 killed-per-accident ratios: Ahmedabad 0.29, Vadodara 0.38, Surat 0.43. 0.33 is a representative value.

Are these values calibrated on Gujarat data?
The formulas are standard; the overall level is calibrated to MoRTH city totals. Individual factors like the turn values are assumptions from the literature, not estimated from local crash data.

9 Traffic-flow and scoring formulas

Chinta 1 min · running time 6:45–7:45

ON THE SLIDE

- Six cards: BPR travel time, Webster signal delay, time-of-day risk, pedestrian fatality, calibration to MoRTH, safety score

FORMULAS ON THIS SLIDE

1. BPR travel time (Bureau of Public Roads, 1964)

$$t = t_0 \left[1 + 0.15 \left(\frac{V}{C}\right)^4\right]$$

t_0 = free-flow time; V/C = volume-to-capacity ratio = peak V/C of the road class × hourly demand. At V/C = 1 travel time is 15% higher; above that it rises steeply (4th power), which is how bottlenecks appear.

2. Webster's uniform signal delay (1958)

$$d = \frac{C\,(1-\lambda)^2}{2\,(1-\lambda x)}$$

C = cycle length (90 s), λ = green ratio (0.45), x = degree of saturation. A busy signal (x = 0.9) costs about 23 s.

3. Time-of-day risk (MoRTH 2022, Table 7.3)

$$\mathrm{TOD}(h) = \frac{\text{share of crashes in } h}{\text{share of traffic in } h}$$

Per vehicle-km: 00–06 h ≈ 1.3×, 09–12 h ≈ 0.8×, 21–24 h ≈ 1.16×. Crash shares are MoRTH data; traffic shares are an assumed typical urban daily profile.

4. Pedestrian fatality (Rosén & Sander, 2009)

$$P(v) = \frac{1}{1 + e^{\,6.9 - 0.09\,v}}$$

Probability a pedestrian hit at v km/h dies: 1.5% at 30, 3.6% at 40, 8.3% at 50, 18% at 60.

5. Calibration

$$k = \frac{N_{\mathrm{MoRTH}}}{\sum_e \mathrm{exposure}_e}$$

One constant k scales every road's crash rate so the predicted yearly total for the city equals the official MoRTH total (Surat 683; Ahmedabad region 1,882 including Gandhinagar).

6. Safety score (our route index)

$$\mathrm{Score} = 100\,\exp\left(-\ln\frac{100}{70}\cdot\frac{r}{\bar{r}}\right)$$

r = the route's risk per km; $\bar{r}$ = average risk per km on the city's roads (traffic-weighted). If $r = \bar{r}$ the score is exactly 70; zero risk gives 100; twice the average gives 49.

UNDERSTAND IT FIRST

These formulas turn the safety model into travel times and one readable score. BPR is the standard traffic-flow relation between volume/capacity and travel time.

Calibration is what makes the numbers realistic: without it the risk values would have no absolute meaning. With it, the model predicts exactly as many crashes per year as MoRTH reports for each city.

SCRIPT (fits 1 min)

The safety formulas tell us how dangerous a road is; these six turn that into travel time and a score. Travel time uses the BPR curve: free-flow time times one plus 0.15 V over C to the fourth. So the same trip is slower at 9 am than at 2 am. Signals add Webster's delay; a busy signal costs about 23 seconds.

Time-of-day risk is MoRTH's share of crashes in each 3-hour window divided by the share of traffic. Midnight to 6 am is about 1.3 times riskier per kilometre. For walking we use the pedestrian fatality curve: 8% at 50 km/h, 18% at 60.

Calibration scales everything so predicted crashes equal the MoRTH total, 683 for Surat. Finally, the safety score: 70 is an average road, higher is safer.

IF ASKED

Why the 4th power in BPR?
It is the standard BPR calibration ($\alpha = 0.15$, $\beta = 4$): delay stays small below capacity and grows sharply near and above it.

Is traffic live?
No. Time-dependent travel time uses an assumed hourly demand profile, not live traffic feeds.

10 The risk model, step by step

Arnav 1 min · running time 7:45–8:45

ON THE SLIDE

- Six-step flow: road graph → features → risk per edge → calibrate → add blackspot history → safety score
- Key numbers: 2.3 lakh junctions, 6.0 lakh directed road links, 4 regions / 5 cities / 23,273 km, 3 modes
- Statement: statistical model calibrated to official totals, not a trained machine-learning model

UNDERSTAND IT FIRST

Graph: junctions are nodes; each road segment becomes two directed edges (one per direction), so one-ways are respected. Only the largest connected part of the network is kept (over 98.9% of nodes in every city), so any start can reach any destination.

Risk per edge = crash frequency for the road class × length × severity S(v) × time-of-day factor × lighting factor × other CMFs × blackspot multiplier. Junctions have their own risk based on number of arms, signal control and the turn being made.

Personal security is also included at night (lighting, roadside activity, police distance, CCTV), with more weight for walking than for cars.

Why not machine learning? Training a crash model needs crash records with exact locations. Those are not public for Gujarat (MoRTH's iRAD system is not open). So we used a statistical model with published factors, calibrated to official totals. This is the honest answer if asked.

SCRIPT (fits 1 min)

Here is how the theory becomes numbers the computer can route on. First, OpenStreetMap roads become a directed graph: junctions are nodes, and each road segment is an edge in each allowed direction. Second, every edge gets its features: speed, road class, lighting, bridges, bus stops.

Third, risk per edge is crash frequency times the severity term times the time-of-day factor, with the lighting and road factors shown earlier. Junctions get their own risk depending on the turn. Fourth, we calibrate so each city's predicted crashes equal the MoRTH total. Fifth, Empirical Bayes raises risk at the 22 blackspots. Finally each route gets a score from 0 to 100.

To be precise: this is a calibrated statistical model, not a trained ML model, because geo-coded crash data is not public for Gujarat.

IF ASKED

Why not train an ML model?
No public crash records with locations for these cities. With iRAD data we would train a crash-frequency model; that is a next step.

How big is the network?
About 2.3 lakh junctions and 6 lakh directed links across 23,273 km.

11 Routing engine: turn-aware A* search

Arnav 1 min 10 s · running time 8:45–9:55

ON THE SLIDE

- The route cost formula
- Four design points: turn-aware search, guaranteed optimal, detour cap, Pareto front → Balanced route
- Every query returns Fastest, Balanced and Safest; checks: A* matched Dijkstra 48/48; about 0.25 s per full plan

FORMULAS ON THIS SLIDE

$$C = \lambda \frac{R}{R_0} + (1 - \lambda) \frac{T}{T_0}$$

R = route risk, T = route time; R_0, T_0 = risk and time of the fastest route between the same two points; λ = the user's safety priority (0 = only speed, 1 = only safety). Dividing by the fastest route makes risk and time unit-free percentages, so they can be added.

UNDERSTAND IT FIRST

A* is Dijkstra's shortest-path algorithm plus a heuristic that estimates the remaining cost to the destination, so it explores fewer roads. If the heuristic never overestimates (“admissible”), A* still finds the optimal route. Ours uses straight-line distance at maximum speed plus the minimum possible risk per metre, which can never overestimate.

Turn-aware: the search state is “which road did I arrive on”, not just “which junction am I at”. That lets every manoeuvre (left, straight, right, U-turn) have its own cost. Without it, the “safest” route zig-zags through many lanes.

Detour cap: if the safest route is more than 25% slower than the fastest, λ is lowered step by step (bisection) until the route fits the time budget.

Pareto front and Balanced route: solving for several λ values gives a set of routes where none is both faster and safer than another. The Balanced route is the “knee” of that curve: the most risk removed per extra minute.

SCRIPT (fits 1 min 10 s)

Risk and time have different units, so we divide each by the fastest route's own value. Both become percentages of the normal trip, and λ, the safety slider, sets the trade-off.

Four points. One: we search over turns, not just roads, so a right turn across traffic costs more than going straight. Two: it is provably optimal. Our heuristic never overestimates, so A* finds the true best route; we checked it against Dijkstra, and 48 out of 48 matched. Three: a detour cap. If the safest route is more than 25% slower, we lower λ until it fits. Four: we compute the trade-off curve and pick its knee as the Balanced route.

Every query returns Fastest, Balanced and Safest in about a quarter of a second.

IF ASKED

Why A* and not Dijkstra?

Same optimal answer, verified 48/48. A* explores fewer nodes thanks to the admissible heuristic, and ours also costs turns.

How do you combine risk and time?
Divide each by the fastest route's value, so both are percentages of the normal trip, then weight with λ.

12 System architecture

Veer 50 s · running time 9:55–10:45

ON THE SLIDE

- Four stages: data pipeline (Python) → prebuilt graphs (NumPy files) → backend API (FastAPI) → web interface (MapLibre + JavaScript)
- 57 automated tests passing; 400 simulated trips in validation; open data only

UNDERSTAND IT FIRST

The pipeline runs once and builds a risk graph per city in about a minute; the result is about 28 MB for all cities. After that, routing needs no live service: if an outside API fails, routing still works.

API endpoints: route (plan a trip), search (find places, even acronyms like “svnit”), reverse (name a clicked point), layers (blackspots, police, hospitals), network (risk heatmap). Every input is checked: bad coordinates, impossible hours or a city outside our five return a clear error message instead of crashing.

The 57 tests cover the formulas, A* optimality, API input checks and HTTP behaviour.

SCRIPT (fits 50 s)

How the pieces connect. A Python data pipeline runs once: it downloads the data, locates the blackspots and builds a risk graph for each city in about a minute. The output is a set of NumPy files, about 28 MB in total, so once built, routing needs no live service.

The backend is FastAPI: it plans routes, searches places, serves blackspots, police and hospitals, and the risk heatmap. Every input is validated, so bad requests get a clear error instead of a crash. The web interface uses MapLibre for the map. 57 automated tests pass, and all data is open and cited.

IF ASKED

What if an API goes down?
Routing runs entirely on the prebuilt local graphs. Only map tiles and fallback place search need the internet.

13 The interface

Veer 1 min · running time 10:45–11:45

ON THE SLIDE

- Screenshot: SVNIT → Surat Railway Station by car at 9 pm
- Five features: search/click/locate; mode, time and priority controls; three route cards; explanations; mobile, dark mode, shareable links

UNDERSTAND IT FIRST

The goal was a Google-Maps-like page that answers a different question. Route colours: Safest blue, Balanced green, Fastest orange. The selected route is coloured along its length by risk level (low to very high).

Each card shows travel time, extra minutes vs fastest, % lower risk and the 0–100 score. Tabs show where the risk comes from, blackspots on the route, turn-by-turn directions with risk per step, and a risk-along-route chart.

If there is time, a short live demo here is the most convincing part of the talk.

SCRIPT (fits 1 min)

This is the real interface, planning SVNIT to Surat Railway Station at 9 pm. In the left panel you pick a city, search or click start and destination, choose car, two-wheeler or walking, and set the departure time, because risk changes with the hour. The safety slider is λ from the cost function, and you can set a maximum detour.

You get up to three cards: Safest in blue, Balanced in green, Fastest in orange. Each shows time, extra minutes, percentage lower risk and a safety score, and the route is coloured by risk along its length. Tabs explain the choice: blackspots passed, risky turns and a risk-along-route chart. It also works on mobile and in dark mode.

IF ASKED

Can it run on a phone?
Yes, it has a mobile bottom-sheet layout; it is not yet hosted publicly (a next step).

14 Case study: SVNIT → Surat Station

Chinta 55 s · running time 11:45–12:40

ON THE SLIDE

- Table (car, 9 pm): Fastest 22.5 min, score 66; Balanced 25.0 min (+2.5), score 76, −37% risk; Safest 28.0 min (+5.5), score 80, −46% risk
- Why they differ: the fastest uses the flyover chain (91% on arterials); Balanced uses Nanawat Main Road (42% on arterials)
- Survey meets model: 57.9% would accept 2–3 extra minutes; Balanced costs +2.5 min

UNDERSTAND IT FIRST

The fastest route runs over the Majura Gate, Udhna Darwaja and Ring Road flyovers: high speed, so high severity per crash. Respondents also named the Ring Road flyover stretch as a congestion point.

The same pattern holds at 9 am and for two-wheelers: 40–48% lower modelled risk for the safest route.

These are model outputs, not observed crashes. Field validation (timing the routes and a safety audit) is planned.

SCRIPT (fits 55 s)

Here the survey and the model meet, on the same corridor our respondents travel. By car at 9 pm, the fastest route takes 22.5 minutes with a score of 66. The Balanced route takes 2.5 minutes more, scores 76, with 37% lower modelled risk. The Safest takes 5.5 minutes more with 46% lower risk.

Why? The fastest route stays on the high-speed flyover chain, 91% on arterials. The Balanced route goes through Nanawat Main Road, only 42% on arterials, so speeds and crash severity are lower. And the link to our survey: 57.9% said they'd accept 2 to 3 extra minutes; the Balanced route costs exactly 2.5.

IF ASKED

Have you verified these times on the road?
Not yet; they are model estimates. Timing Routes A, B and C in the field is the first next step.

15 Validation: 400 simulated trips

Arnav 50 s · running time 12:40–13:30

ON THE SLIDE

- Chart: average risk reduction, safest vs fastest, by city at 9 am and 10 pm
- Callouts: −26.9% average modelled risk, +3.5 min average extra time, blackspot passes 107 → 41, average score 70 → 78
- Night gains are larger

UNDERSTAND IT FIRST

Method: in each city, 25 random origin–destination pairs 3–15 km apart, each planned for car and two-wheeler at 9 am and 10 pm = 400 trips, default settings (priority 0.7, 25% detour cap). The random seed is fixed so results are reproducible.

By city (9 am → 10 pm): Ahmedabad 25.5% → 28.4%, Surat 23.8% → 25.6%, Vadodara 24.1% → 29.3%, Rajkot 25.5% → 33.4%.

Honest finding (in the documentation): at night the safety route sometimes carries slightly more poorly-lit distance, because it prefers slower local roads and the satellite light data cannot separate neighbouring streets.

SCRIPT (fits 50 s)

To test beyond one corridor we ran 400 simulated trips: 25 random trips per city, for car and two-wheeler, at 9 am and 10 pm. On average the safest route had 26.9% lower modelled risk for 3.5 extra minutes. The average score rose from 70 to 78, and passes through documented blackspots fell from 107 to 41.

In every city the gain is larger at night. Rajkot goes from 25.5% by day to 33.4% at night. These are modelled risks; real validation needs geo-coded crash data, which is our next step.

IF ASKED

Is this proof that the routes are safer?
No. It shows the method consistently finds lower-risk routes under the model. Proof needs observed crash data.

16 Limitations and next steps

Akshara 40 s · running time 13:30–14:10

ON THE SLIDE

- What we can't claim yet: risk is modelled, blackspot data is uneven, satellite lighting is coarse, survey represents SVNIT commuters, no inter-city routing (AQI belongs to the other team)
- Next steps: field validation, data-driven crash model, deployment, integration with AQI and efficiency

UNDERSTAND IT FIRST

Being open about limits is a strength in a viva; every limitation here has a matching next step.

iRAD (Integrated Road Accident Database) is MoRTH's crash database with locations; access would allow a trained crash-frequency model.

SCRIPT (fits 40 s)

We want to be clear about what this work does not show. The risk numbers are modelled, calibrated to official totals but not observed street by street. Blackspot data is uneven, satellite lighting is coarse, and the survey represents SVNIT commuters.

Next: field validation, timing Routes A, B and C and auditing the corridor. A data-driven crash model once geo-coded iRAD data is available. A hosted, phone-friendly demo. And integration with the AQI and efficiency objectives.

17 Thank you

Chinta 20 s · running time 14:10–14:30

ON THE SLIDE

- Summary: 5 cities, one safety router; −27% modelled risk; +3.5 min average cost; 22 blackspots mapped
- Team and supervisor

UNDERSTAND IT FIRST

Keep it short; then open for questions and offer the live demo.

SCRIPT (fits 20 s)

To summarise: one safety router for five Gujarat cities, built on public data and established traffic-safety formulas. On average it cuts modelled risk by about 27% for 3.5 extra minutes. Thank you. We are happy to take questions, and we can show the live demo.

Questions the panel may ask (anyone)

Why A* and not Dijkstra?
Same optimal route; we verified 48 of 48 cases. A* explores fewer roads thanks to a heuristic that never overestimates, and ours also costs turns.

How do you add risk and time when they have different units?
We divide each by the fastest route's own risk and time, so both become percentages of the normal trip, then weight them with the safety slider λ.

Where is the crash data from?
MoRTH Road Accidents in India 2022 for city totals and time of day; 22 blackspots from the Gujarat Road Safety Authority, the State Traffic Branch and MoRTH's highway blackspot list.

Is this machine learning?
No. It is a statistical model built from published factors and calibrated to official totals. Training an ML model needs geo-coded crash records, which are not public for Gujarat.

What if an online API fails?
Routing runs fully on prebuilt local graphs. Only map tiles and fallback place search need the internet.

How do you know the safe route is really safer?
Today it is a model-based comparison. Field timing, a road-safety audit of the corridor and access to iRAD crash data are how we would validate it.

Why does the safe route sometimes use small roads?
Lower speeds sharply reduce severity (power model), and the turn penalty prevents zig-zag routes. The detour cap keeps the extra time reasonable.

What is the safety score?
A 0–100 index of risk per km compared with the city's average road. 70 = average, higher = safer.

Key numbers cheat sheet

Item	Value
Deaths 2022 (Ahd + Surat + Rajkot + Vadodara)	488 + 293 + 150 + 181 = 1,112
Crashes between 18:00 and 21:00 (India)	20.4%
Survey	100 responses; 57.9% accept 2–3 min extra (n = 95); 45.1% speeding at junctions (n = 91); 31% Station Approach Road (n = 100); 67.4% face 5–15 min delay (n = 95)
Network	23,273 km; about 2.3 lakh junctions; about 6.0 lakh directed links; 4 regions, 5 cities
Blackspots	22 (Ahmedabad 15, Vadodara 4, Gandhinagar 1, Surat 1, Rajkot 1); Jambuva bridge 36 crashes, 30 deaths, weight ×11.3
Case study (car, 9 pm)	Fastest 22.5 min, score 66 · Balanced 25.0 min, 76, −37% · Safest 28.0 min, 80, −46%
Validation (400 trips)	−26.9% modelled risk, +3.5 min, blackspot passes 107 → 41, score 70 → 78

Engine checks	A* = Dijkstra 48/48; about 0.25 s per plan; 57 automated tests